Procurement Recommendation: Supplier Delivery-Risk & Exposure

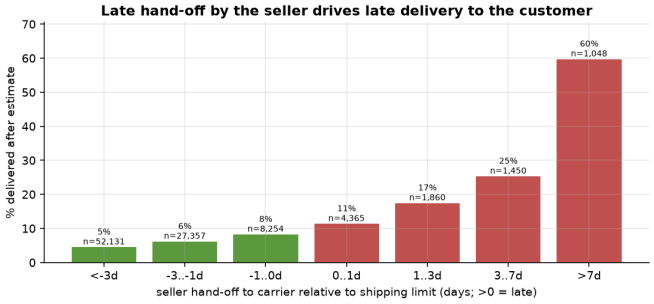
To: Head of Procurement / Marketplace Ops | **From:** Supplier Analytics | **Data:** Olist Brazilian E-Commerce, purchases 2016-09-04 to 2018-09-03 (112,650 items, 3,095 sellers). Every figure is computed by the pipeline in this repository. **Olist sellers are used as proxies for suppliers.** Items marked **ASSUMED** are scenario inputs, not observed results.

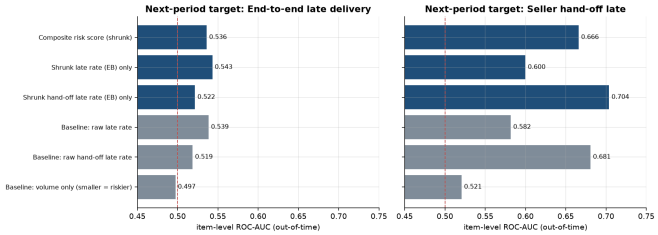
1. Empirical-Bayes seller scoring to predict dispatch delays across 3,095 Olist sellers

Problem: 6.8% of delivered orders arrived late; which sellers drive delivery risk and working-capital exposure? **Method:** SQL delivery intervals, beta-binomial shrinkage with bootstrap intervals, out-of-time validation vs baselines, cold-start policy, indicative archetypes, R GLM/GLMM. **Decision:** fix dispatch, not volume (section 6).

2. What the data says (observed)

- **6.8% of 96,470 delivered orders arrived after the promised date** (mean 10.6 days late). Late orders average a 2.27/5 review vs 4.29 on time.
- **The seller-controlled leg matters most.** 9.4% of items were handed to the carrier after the shipping limit; those orders were late 21% of the time vs 5.4% otherwise. Controlling for lane, region, category, freight and season, a late hand-off multiplies the odds of a late delivery by **4.95** (95% CI 4.65- 5.28; logistic GLM, R).
- **But end-to-end lateness is mostly not a seller trait.** In a seller random-intercept model only 8.8% of latent variance sits between sellers (5.8% after hand-off is controlled). Platform-wide spikes (Nov-2017, Feb-Mar 2018) dominate.
- **Spend is concentrated:** the top 10% of sellers hold 68% of GMV and 71% of GMV-at-risk.





3. Does the score predict next period? (out-of-time, scored at 2018-01-01)

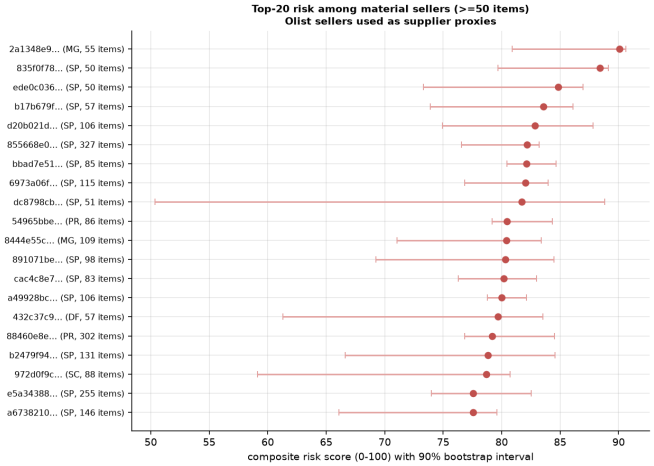
1,081 sellers scored on pre-2018 data, evaluated on items bought 2018-01-01 to 2018-09-01 (75% of next-period items).

Score	AUC: next hand-off late	Top-10% lift	AUC: next late delivery
EB-shrunk hand-off rate	**0.704**	2.26x	0.522
Composite risk score	0.666	2.36x	0.536
Raw hand-off rate (baseline)	0.681	2.56x	0.519
Raw late rate (baseline)	0.582	2.17x	0.539
Volume only (baseline)	0.521	1.30x	0.497

- Shrinkage beats the raw rate on hand-off lateness (dAUC +0.023, 95% CI +0.012 to +0.039) and halves the error for sellers with <5 deliveries (MAE 0.124 -> 0.058).
- The riskiest score decile had a next-period hand-off late rate of 21.0% vs 2.9% in the safest. **No score predicts end-to-end lateness well** (AUC <= 0.543); it is carrier/lane/calendar driven.
- **Cold start:** 1,249 new sellers (24.8% of next-period items) had no history; policy assigns the EB prior mean and a **New / unscored** tier. Their next hand-off late rate was 8.7% vs a 12.7% prior (conservative for dispatch).

4. Who to act on

- **69 material sellers (>=50 items) are in the High/Elevated tiers**, holding 7.6% of GMV and 13.7% of late GMV. The full High tier (310 sellers) carries 6.5% of GMV but 17.5% of late GMV, yet its median seller has only 4 items - treat these as *unknown*, not *bad*.
- **"Slow dispatchers" (indicative archetype):** 144 sellers, 10.5% of GMV, median hand-off late rate 25.8% vs 4.8% for core suppliers - the priority group for dispatch-SLA conversations. (Indicative only: silhouette 0.174.)





5. Exposure and the reallocation scenario

- Observed: GMV BRL 13.59m, late GMV BRL 986k, GMV-at-risk BRL 914k, freight BRL 2.25m. Working-capital proxy: BRL 407k of GMV in transit on average; delays add BRL 30k.
- **ASSUMED** costs (5% of GMV per late item, 30% freight absorbed, 10% per cancellation) give total exposure of BRL 729k - a scenario figure, not a P&L.
- **Back-tested reallocation (ASSUMED):** moving 50% of the top-10% sellers' next-period volume, closing 50% of the gap, would have avoided only 0.29% of late GMV (BRL 1,282); even the most aggressive grid cell (top 20%, all volume, full gap) gives 1.7%. On the seller-controlled leg the same move cuts late hand-offs by 4.1%.

6. Recommendations & trade-offs

1. **Adopt the EB-shrunk hand-off late rate as the supplier SLA KPI**, reviewed monthly with intervals; it is the most predictive, least noisy signal and is what sellers control.
2. **Run dispatch-SLA reviews with the 69 material High/Elevated sellers**, starting with "Slow dispatchers"; use the composite only for triage (it adds review/cancellation/freight context but predicts no better).
3. **Do not expect volume reallocation to fix customer lateness**; carrier and lane performance (North/Northeast lanes, peak months) need a logistics workstream.
4. **Onboarding watch-list, not penalties, for low-volume sellers** (1,201 have <5 items).

Limitations: sellers are marketplace merchants, not contracted suppliers; costs are ASSUMED; scores cover 75% of next-period items (new sellers unscored); multi-seller orders share timestamps; GLM effects are associations. Data: Olist, CC BY-NC-SA 4.0.